Notes on Kalman filter for PTA analysis

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This note defines the Kalman "machinery" for the state-space formulation of PTA analysis in terms of timing residuals. It is heavily based on work by P. Meyers and the MINNOW package, with extensions to handle multiple pulsars and the influence of the stochastic GW background, with contributions from A. Vargas. This work builds on our previous papers, continuing from their state-space formulation in the frequency domain.

1 Introduction

For the purposes of running state-space algorithms, we define the following:

- $\boldsymbol{X}$: the state vector, dimension n_X
- $\boldsymbol{Y}$: the observation vector, dimension n_Y
- $\boldsymbol{F}$: the state transition matrix, dimension $n_X \times n_X$
- $\boldsymbol{H}$: the measurement matrix, dimension $n_Y \times n_X$
- $\boldsymbol{Q}$: the process noise covariance matrix, dimension $n_X \times n_X$
- $\boldsymbol{R}$: the measurement noise covariance matrix, dimension $n_Y \times n_Y$

The new goal of this note is to define the above matrices.

2 System definition

We have a set of N pulsars. Each pulsar has $4 + M^{(n)}$ states, where $M^{(n)}$ is the number of timing model parameters for the n-th pulsar. The states evolve according to the following set of dynamical equations.

$$\begin{aligned}
\frac{d}{dt}\delta\phi^{(n)} &= \delta f^{(n)}, && (1)\\
\frac{d}{dt}\delta f^{(n)} &= -\gamma_p^{(n)}\delta f^{(n)} + \chi_p^{(n)}(t), && (2)\\
\frac{d}{dt}r^{(n)} &= a^{(n)}, && (3)\\
\frac{d}{dt}a^{(n)} &= -\gamma_a a^{(n)} + \chi_{\mathrm{a}}^{(n)}(t), && (4)\\
\frac{d}{dt}\delta\epsilon_m^{(n)} &= \chi_{\epsilon_m}^{(n)}(t) \quad \forall m \in [1, M^{(n)}]. && (5)
\end{aligned}$$

where χ_i denotes a white noise stochastic process and γ_i are daming constants. The processes $\chi_p^{(n)}$ and $\chi_\epsilon^{(n)}$ are uncorrelated between different n. Their statistics are defined by

$$\begin{aligned}
\langle \chi_p^{(n)}(t)\rangle &= 0\ , && (6)\\
\langle \chi_p^{(n)}(t)\chi_p^{(n')}(t')\rangle &= [\sigma_p^{(n)}]^2\delta_{n,n'}\delta(t-t')\ . && (7)
\end{aligned}$$

The factor $\delta_{n,n'}$ in the right-hand side ensures that the non-GW-induced spin fluctuations in distinct pulsars $n \neq n'$ are uncorrelated, as expected physically.

$$\begin{aligned}
\langle \chi_{\epsilon_m}^{(n)}(t)\rangle &= 0\ , && (8)\\
\langle \chi_{\epsilon_m}^{(n)}(t)\chi_{\epsilon'_m}^{(n')}(t')\rangle &= [\sigma_{\epsilon_m}^{(n)}]^2\delta_{n,n'}\delta(t-t')\ . && (9)
\end{aligned}$$

Note that I am not sure if it makes sense for the timing model parameter errors to be treated in this way; perhaps there is some correlation between the parameters for a single pulsar . For now we just assume that they are all uncorrelated for simplicity.

The correlation between pulsars manifests due to the GW, i.e. the $a^{(n)}$ state. The ensemble statistics of $\chi_{\mathrm{a}}^{(n)}$ are

$$\begin{aligned}
\langle \chi_{\mathrm{a}}^{(n)}(t)\rangle &= 0\ , && (10)\\
\langle \chi_{\mathrm{a}}^{(n)}(t)\,\chi_{\mathrm{a}}^{(n')}(t')\rangle &= \left[\sigma_{\mathrm{a}}^{(n,n')}\right]^2\delta(t-t')\ . && (11)
\end{aligned}$$

with

$$\left[\sigma_{\mathrm{a}}^{(n,n')}\right]^2 = \frac{\langle h^2\rangle}{6}\gamma_{\mathrm{a}}\,\Gamma\left[\theta^{(n,n')}\right]\ , \tag{12}$$

where $\langle h^2\rangle$ is the mean square GW strain, $\theta^{(n,n')}$ is the angle between the n-th and n'-th pulsars, and one has

$$\Gamma\left[\theta^{(n,n')}\right] = \frac{3}{2}x_{nn'}\ln x_{nn'} - \frac{x_{nn'}}{4} + \frac{1}{2} + \frac{1}{2}\delta_{nn'}\ , \tag{13}$$

with $x_{nn'} = \left[1 - \cos\theta^{(n,n')}\right]/2$.

The states described by the SDEs are related to the observables, $\boldsymbol{Y} = (\delta t)$ via a measurement equation

$$\delta t^{(n)} = \frac{\delta\phi^{(n)}}{f_0^{(n)}} + \mathbf{M}^{(n)}\boldsymbol{\delta\epsilon}^{(n)} - r^{(n)} \tag{14}$$

where $\mathbf{M}^{(n)}$ is the design matrix of partial derivatives of the phases with respect to the timing model parameters. It is provided by standard pulsar timing libraries such as TEMPO and is a (row)vector of dimension $M^{(n)}$ (note the clumsy notation; $\mathbf{M}^{(n)}$ is a vector, the normal $M^{(n)}$ is a scalar which sets the dimension of $\mathbf{M}^{(n)}$) The quantity $\boldsymbol{\delta\epsilon}^{(n)}$ is a (column) vector, also of length $M^{(n)}$, which holds all the $\delta\epsilon_m^{(n)}$ terms for pulsar n. The quantity $f_0^{(n)}$ is the nominal spin frequency of pulsar n.

3 Matrix representation

We can represent the system in matrix form as follows.

4 Summary of the Full System

4.1 State-Space Model in Continuous Time

We collect all pulsars' states into one large state vector $\mathbf{X}(t)$. Recall that we have N pulsars, each with the following individual states:

$$\left\{\,\delta\phi^{(n)},\ \delta f^{(n)},\ r^{(n)},\ \delta\epsilon_1^{(n)}, \ldots, \delta\epsilon_{M^{(n)}}^{(n)}\right\}, \tag{15}$$

and we also have the N *GW-correlated* rate states $\{\,a^{(n)}\}_{n=1,\ldots,N}$. We group the states as follows:

$$\mathbf{X}(t) = \begin{pmatrix} a^{(1)}(t) \\ \vdots \\ a^{(N)}(t) \\ \delta\phi^{(1)}(t) \\ \delta f^{(1)}(t) \\ r^{(1)}(t) \\ \delta\epsilon_1^{(1)}(t) \\ \vdots \\ \delta\epsilon_{M^{(1)}}^{(1)}(t) \\ \delta\phi^{(2)}(t) \\ \delta f^{(2)}(t) \\ r^{(2)}(t) \\ \delta\epsilon_1^{(2)}(t) \\ \vdots \\ \delta\epsilon_{M^{(2)}}^{(2)}(t) \\ \vdots \\ \delta\phi^{(N)}(t) \\ \delta f^{(N)}(t) \\ r^{(N)}(t) \\ \delta\epsilon_1^{(N)}(t) \\ \vdots \\ \delta\epsilon_{M^{(N)}}^{(N)}(t) \end{pmatrix}, \tag{16}$$

where:

$$\mathbf{a}(t) = \big(a^{(1)},\, a^{(2)}, \ldots, a^{(N)}\big)^T, \tag{17}$$

$$\mathbf{x}_n(t) = \begin{pmatrix} \delta\phi^{(n)}(t) \\ \delta f^{(n)}(t) \\ r^{(n)}(t) \\ \delta\epsilon_1^{(n)}(t) \\ \vdots \\ \delta\epsilon_{M^{(n)}}^{(n)}(t) \end{pmatrix}. \tag{18}$$

Hence, the dimension of $\mathbf{a}(t)$ is N, and the dimension of each $\mathbf{x}_n(t)$ is $3 + M^{(n)}$, giving a total state-vector size of:

$$\dim \mathbf{X} = N + \sum_{n=1}^{N} \big(3 + M^{(n)}\big). \tag{19}$$

Continuous-Time Dynamics. We write the system of stochastic differential equations (SDEs) in matrix form as

$$\frac{d\mathbf{X}(t)}{dt} \;=\; \mathbf{A}\,\mathbf{X}(t) \;+\; \mathbf{\Gamma}\,\mathbf{w}(t), \tag{20}$$

where:

- $\mathbf{A}$ is a *block* matrix encoding the linear drift terms.
- $\mathbf{\Gamma}$ is a *block* matrix mapping the driving white-noise processes $\mathbf{w}(t)$ into the state equations.
- $\mathbf{w}(t)$ is a vector of white-noise processes whose covariance may have both diagonal (uncorrelated) parts and block off-diagonal parts (specifically for the GW-correlated $a^{(n)}$).

Block Structure of A

Partition $\mathbf{X}(t)$ as:

$$\mathbf{X}(t) \;=\; \begin{pmatrix} \mathbf{a}(t) \\ \mathbf{x}_1(t) \\ \mathbf{x}_2(t) \\ \vdots \\ \mathbf{x}_N(t) \end{pmatrix}, \quad \text{where} \quad \mathbf{a}(t) \in \mathbb{R}^N, \quad \mathbf{x}_n(t) \in \mathbb{R}^{3+M^{(n)}}. \tag{21}$$

Then $\mathbf{A}$ naturally decomposes into sub-blocks:

$$\mathbf{A} \;=\; \begin{pmatrix} -\gamma_a\, I_N & 0 & 0 & \cdots & 0 \\ C_1 & A_1 & 0 & \cdots & 0 \\ C_2 & 0 & A_2 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ C_N & 0 & 0 & \cdots & A_N \end{pmatrix}. \tag{22}$$

Here:

- The top-left block $-\gamma_a\, I_N$ corresponds to

$$\dot{a}^{(n)}(t) \;=\; -\gamma_a\, a^{(n)}(t),$$

  which is the damping term in the GW-correlated rates (ignoring noise for the moment).

- Each diagonal block A_n is a $(3+M^{(n)}) \times (3+M^{(n)})$ matrix governing the local pulsar states:

$$A_n = \begin{pmatrix} 0 & 1 & 0 & 0 & \cdots & 0 \\ 0 & -\gamma_p^{(n)} & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 0 \end{pmatrix},$$

where the rows/columns are indexed as

$$(\delta\phi^{(n)},\ \delta f^{(n)},\ r^{(n)},\ \delta\epsilon_1^{(n)}, \ldots, \delta\epsilon_{M^{(n)}}^{(n)}).$$

Hence:

$$\begin{cases} \dfrac{d}{dt}\,\delta\phi^{(n)} = \delta f^{(n)}, \\ \dfrac{d}{dt}\,\delta f^{(n)} = -\gamma_p^{(n)}\,\delta f^{(n)}, \\ \dfrac{d}{dt}\,r^{(n)} = 0 \quad \text{(in the drift)}, \\ \dfrac{d}{dt}\,\delta\epsilon_m^{(n)} = 0 \quad \text{(in the drift)}, \end{cases}$$

with no coupling among these pulsar-specific states *except* via noise and the $a^{(n)}$ block below.

- Each C_n is a $(3+M^{(n)}) \times N$ matrix inserting the effect of $\mathbf{a}(t)$ onto $\dot{\mathbf{x}}_n(t)$. Specifically,

$$\dot{r}^{(n)}(t) = a^{(n)}(t),$$

so in the row corresponding to $r^{(n)}$, there is a '1' in column n (and zeros elsewhere). Concretely,

$$C_n = \begin{pmatrix} 0 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & & \vdots \end{pmatrix} \quad \text{(with '1' in row } r^{(n)}\text{, column } n\text{)},$$

and all other entries zero.

Noise-Injection Matrix $\boldsymbol{\Gamma}$

In the SDE,

$$\frac{d\mathbf{X}(t)}{dt} = \mathbf{A}\,\mathbf{X}(t) + \boldsymbol{\Gamma}\,\mathbf{w}(t),$$

the matrix $\mathbf{\Gamma}$ maps the driving noise processes $\mathbf{w}(t)$ into the state. We typically collect all white-noise processes into a single vector:

$$\mathbf{w}(t) = \big(\chi_a^{(1)}(t), \ldots, \chi_a^{(N)}(t),\ \chi_p^{(1)}(t), \chi_{\epsilon_1}^{(1)}(t), \ldots, \chi_{\epsilon_{M^{(1)}}}^{(1)}(t),\ \ldots,\ \chi_p^{(N)}(t), \chi_{\epsilon_1}^{(N)}(t), \ldots\big)^T.$$

We then arrange $\mathbf{\Gamma}$ in a block-diagonal style, consistent with $\mathbf{A}$:

$$\mathbf{\Gamma} = \begin{pmatrix} \Gamma_a & 0 & \cdots & 0 \\ 0 & \Gamma_1 & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & \Gamma_N \end{pmatrix},$$

where

- Γ_a is $N \times N$, typically the identity I_N if we prefer to keep the correlation of the $a^{(n)}$ noises in the covariance of $\chi_a^{(n)}$;
- each Γ_n is $(3 + M^{(n)}) \times (\,1 + M^{(n)}\,)$ (one dimension for $\chi_p^{(n)}$ plus $M^{(n)}$ for $\chi_{\epsilon_m}^{(n)}$) and places those noise terms in the correct rows:

$$\begin{cases} \delta\dot{f}^{(n)}(t) \ \leftarrow\ \chi_p^{(n)}(t), \\ \delta\dot{\epsilon}_m^{(n)}(t) \ \leftarrow\ \chi_{\epsilon_m}^{(n)}(t). \end{cases}$$

4.2 Measurement Equation

Let

$$\mathbf{Y}(t) = \begin{pmatrix} \delta t^{(1)}(t) \\ \delta t^{(2)}(t) \\ \vdots \\ \delta t^{(N)}(t) \end{pmatrix}.$$

Each pulsar's measurement $\delta t^{(n)}(t)$ is given by

$$\delta t^{(n)}(t) \ = \ \frac{\delta\phi^{(n)}(t)}{f_0^{(n)}} \ - \ r^{(n)}(t) \ + \ \mathbf{M}^{(n)}\, \boldsymbol{\delta\epsilon}^{(n)}(t),$$

where $f_0^{(n)}$ is the nominal spin frequency and $\mathbf{M}^{(n)}$ is the (row) design matrix for pulsar n. In matrix form,

$$\mathbf{Y}(t) \ = \ \mathbf{H}\,\mathbf{X}(t), \tag{23}$$

where $\mathbf{H}$ is $N \times \big(N + \sum_n (3 + M^{(n)})\big)$, but it has a block-sparse structure: the n-th row of $\mathbf{H}$ picks out $\frac{1}{f_0^{(n)}}\,\delta\phi^{(n)}$, $-r^{(n)}$, and $\mathbf{M}^{(n)}\,\delta\epsilon^{(n)}$ from the n-th block of $\mathbf{X}(t)$, with *zeros* everywhere else (including the GW-rate block $\mathbf{a}$).

4.3 Continuous-to-Discrete Transformation

In practice, pulsar timing data $\delta t^{(n)}$ are sampled at discrete epochs t_k. The *Kalman filter* typically requires a discrete-time state-update relation:

$$\mathbf{X}(t_{k+1}) \;=\; \boldsymbol{\Phi}\big(\Delta t_k\big)\,\mathbf{X}(t_k) \;+\; \boldsymbol{\eta}(t_k),$$

where $\Delta t_k = t_{k+1} - t_k$, and $\boldsymbol{\eta}(t_k)$ encapsulates the integrated effect of the continuous-time noise from t_k to t_{k+1}. For a linear SDE (20), the *exact* state-transition matrix is

$$\boldsymbol{\Phi}\big(\Delta t_k\big) \;=\; \exp\!\big(\mathbf{A}\,\Delta t_k\big).$$

Then the next state is given by

$$\mathbf{X}(t_{k+1}) \;=\; \exp\!\big(\mathbf{A}\,\Delta t_k\big)\,\mathbf{X}(t_k) \;+\; \int_0^{\Delta t_k} \exp\!\big(\mathbf{A}\,(\Delta t_k - \tau)\big)\,\boldsymbol{\Gamma}\,\mathbf{w}(t_k + \tau)\,d\tau. \tag{24}$$

Defining the discrete-time noise increment

$$\boldsymbol{\eta}(t_k) \;=\; \int_0^{\Delta t_k} \exp\!\big(\mathbf{A}\,(\Delta t_k - \tau)\big)\,\boldsymbol{\Gamma}\,\mathbf{w}(t_k + \tau)\,d\tau,$$

we see that $\boldsymbol{\eta}(t_k)$ is a *zero-mean* random variable (assuming $\mathbf{w}(t)$ has zero mean) whose covariance is

$$\mathbf{Q}(\Delta t_k) \;=\; \mathrm{Cov}\big[\boldsymbol{\eta}(t_k)\big] \;=\; \int_0^{\Delta t_k} \exp\!\big(\mathbf{A}\,(\Delta t_k - \tau)\big)\,\boldsymbol{\Gamma}\,\mathbf{Q}_w\,\boldsymbol{\Gamma}^T\,\exp\!\big(\mathbf{A}^T\,(\Delta t_k - \tau)\big)\,d\tau, \tag{25}$$

where $\mathbf{Q}_w$ is the covariance matrix of the white-noise vector $\mathbf{w}(t)$, *including* any cross-pulsar correlations in the $\chi_a^{(n)}$ components.

This mapping (24)–(25) is the *exact* solution for a linear SDE with constant $\mathbf{A}$ and $\boldsymbol{\Gamma}$. In practice, one may use matrix exponentials and (if $\mathbf{A}$ is large and block-structured) possibly exploit the block structure to compute $\exp(\mathbf{A}\,\Delta t_k)$ more efficiently. No additional approximation (like Euler-Maruyama) is strictly needed for the linear-Gaussian case.

4.4 Discrete-Time Measurement

Once discretized, the measurement equation (23) at discrete times t_k remains linear:

$$\mathbf{Y}(t_k) \;=\; \mathbf{H}\,\mathbf{X}(t_k),$$

assuming that the design matrices $\mathbf{M}^{(n)}$ and $f_0^{(n)}$ are constant over the relevant time interval. (In cases where $\mathbf{M}^{(n)}$ changes between epochs, one may simply update $\mathbf{H}$ at each epoch.)

Summary: The pair of equations

$$\begin{cases} \mathbf{X}(t_{k+1}) \;=\; \boldsymbol{\Phi}\big(\Delta t_k\big)\,\mathbf{X}(t_k) \;+\; \boldsymbol{\eta}(t_k), \\ \mathbf{Y}(t_k) \;=\; \mathbf{H}\,\mathbf{X}(t_k), \end{cases}$$

with

$\mathbf{\Phi}(\Delta t_k) = \exp\bigl(\mathbf{A}\,\Delta t_k\bigr), \quad \mathrm{Cov}[\boldsymbol{\eta}(t_k)] = \mathbf{Q}(\Delta t_k), \quad \mathrm{Cov}[\mathbf{Y}(t_k) \mid \mathbf{X}(t_k)] = \mathbf{R}_k$ (if any measurement noise is pr

is the standard form used in *discrete-time* Kalman filtering for this Pulsar Timing Array application.

$$\Phi(\Delta t) = \exp\Big(A\,\Delta t\Big) = \begin{pmatrix} F_{aa} & 0 \\ F_{xa} & F_{xx} \end{pmatrix}. \tag{26}$$

The components are as follows:

(a) The a-block:

$$F_{aa} = \exp\Big(-\gamma_a\,\Delta t\Big) I_N\,. \tag{27}$$

(b) The pulsar-specific blocks: For each pulsar n, the local dynamics block A_n (of size $3 + M^{(n)}$) has the nontrivial 2×2 sub-block for $(\delta\phi^{(n)}, \delta f^{(n)})$

$$\exp\Big(A_n^{(2\times2)}\,\Delta t\Big) = \begin{pmatrix} 1 & \dfrac{1 - e^{-\gamma_p^{(n)}\Delta t}}{\gamma_p^{(n)}} \\ 0 & e^{-\gamma_p^{(n)}\Delta t} \end{pmatrix}, \tag{28}$$

while the remaining states (namely, $r^{(n)}$ and the timing parameters $\delta\epsilon_m^{(n)}$) evolve with zero drift and thus have the identity mapping. Therefore, we define

$$F_n \equiv \exp\Big(A_n \Delta t\Big) = \begin{pmatrix} 1 & \dfrac{1 - e^{-\gamma_p^{(n)}\Delta t}}{\gamma_p^{(n)}} & 0 & \cdots & 0 \\ 0 & e^{-\gamma_p^{(n)}\Delta t} & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ 0 & 0 & 0 & \ddots & 0 \\ 0 & 0 & 0 & \cdots & 1 \end{pmatrix}. \tag{29}$$

Then, collecting all pulsar blocks,

$$F_{xx} = \mathrm{diag}\Big\{F_1, F_2, \ldots, F_N\Big\}\,. \tag{30}$$

(c) The off-diagonal block F_{xa}: This block collects the influence of the a-states on each pulsar's state. For each pulsar n, we have

$$F_{x_n,a} = \int_0^{\Delta t} \exp\Big(A_n(\Delta t - s)\Big)\, C_n\, e^{-\gamma_a s}\, ds\,. \tag{31}$$

Since C_n has only one nonzero entry (a 1 in the third row corresponding to $r^{(n)}$), and the evolution of $r^{(n)}$ is the identity, it follows that for pulsar n

$$F_{x_n,a} = \begin{pmatrix} 0 \\ 0 \\ \int_0^{\Delta t} e^{-\gamma_a s}\, ds \\ 0 \\ \vdots \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \dfrac{1 - e^{-\gamma_a \Delta t}}{\gamma_a} \\ 0 \\ \vdots \\ 0 \end{pmatrix} . \tag{32}$$

Collecting over all pulsars we write

$$F_{xa} = \mathrm{diag}\Big\{ F_{x_1,a}, F_{x_2,a}, \ldots, F_{x_N,a} \Big\} . \tag{33}$$

Thus, the full state-transition matrix is given by

$$\boxed{\Phi(\Delta t) = \begin{pmatrix} e^{-\gamma_a \Delta t}\, I_N & 0 \\ \mathrm{diag}\{F_{x_1,a}, \ldots, F_{x_N,a}\} & \mathrm{diag}\{F_1, \ldots, F_N\} \end{pmatrix} .} \tag{34}$$

The exact process noise covariance is given by

$$Q(\Delta t) = \int_0^{\Delta t} \exp\Big(A(\Delta t - s)\Big)\, \Gamma\, Q_w\, \Gamma^T\, \exp\Big(A^T(\Delta t - s)\Big)\, ds , \tag{35}$$

where Q_w is the covariance matrix of the continuous white-noise vector $w(t)$. With the block structure of A and Γ the matrix $Q(\Delta t)$ partitions as

$$Q(\Delta t) = \begin{pmatrix} Q_{aa} & Q_{a,x} \\ Q_{x,a} & Q_{xx} \end{pmatrix} . \tag{36}$$

(a) The a-block: For the GW-correlated states $a^{(n)}$ with dynamics

$$\frac{d}{dt} a^{(n)} = -\gamma_a\, a^{(n)} + \chi_a^{(n)},$$

the process noise covariance is

$$Q_{aa} = \int_0^{\Delta t} e^{-\gamma_a(\Delta t - s)}\, Q_a\, e^{-\gamma_a(\Delta t - s)} ds = \frac{1 - e^{-2\gamma_a \Delta t}}{2\gamma_a}\, Q_a , \tag{37}$$

where Q_a is the covariance matrix for the χ_a noise (including any cross-pulsar correlations).

(b) The pulsar-specific blocks: For each pulsar n, let Q_n denote the covariance for the pulsar-specific white-noise processes acting on $\delta f^{(n)}$ (with variance $\sigma_p^{(n)2}$) and on $\delta\epsilon_m^{(n)}$ (with variance $\sigma_{\epsilon_m}^{(n)2}$). In pulsar n the contribution is

$$Q_{xx}^{(n)} = \int_0^{\Delta t} \exp\Big(A_n(\Delta t - s)\Big)\,\Gamma_n\, Q_n\, \Gamma_n^T\, \exp\Big(A_n^T(\Delta t - s)\Big) ds\,. \tag{38}$$

For example, focusing on the $(\delta\phi^{(n)}, \delta f^{(n)})$ subsystem where noise $\chi_p^{(n)}$ is injected only into the $\delta f^{(n)}$ equation, we use the 2×2 sub-block

$$\exp\Big(A_n^{(2\times 2)} t\Big) = \begin{pmatrix} 1 & \dfrac{1 - e^{-\gamma_p^{(n)} t}}{\gamma_p^{(n)}} \\ 0 & e^{-\gamma_p^{(n)} t} \end{pmatrix}. \tag{39}$$

Defining the integrals

$$I_1^{(n)} = \int_0^{\Delta t} \left(\frac{1 - e^{-\gamma_p^{(n)} s}}{\gamma_p^{(n)}}\right)^2 ds\,, \tag{40}$$

$$I_2^{(n)} = \int_0^{\Delta t} \frac{1 - e^{-\gamma_p^{(n)} s}}{\gamma_p^{(n)}}\, e^{-\gamma_p^{(n)} s} ds\,, \tag{41}$$

$$I_3^{(n)} = \int_0^{\Delta t} e^{-2\gamma_p^{(n)} s} ds\,, \tag{42}$$

the 2×2 covariance block is

$$Q_{(\delta\phi,\delta f)}^{(n)} = \sigma_p^{(n)2} \begin{pmatrix} I_1^{(n)} & I_2^{(n)} \\ I_2^{(n)} & I_3^{(n)} \end{pmatrix}. \tag{43}$$

For each timing–model parameter $\delta\epsilon_m^{(n)}$, driven by white noise with variance $\sigma_{\epsilon_m}^{(n)2}$, the variance integrated over Δt is simply

$$Q_{\delta\epsilon_m,\delta\epsilon_m}^{(n)} = \sigma_{\epsilon_m}^{(n)2}\, \Delta t\,. \tag{44}$$

Note that the state $r^{(n)}$ has no direct pulsar-specific noise (its noise arises from the coupling with $a^{(n)}$). Thus, for each pulsar n the full $Q_{xx}^{(n)}$ is given by the block-diagonal structure

$$Q_{xx}^{(n)} = \text{block-diag}\Big\{\sigma_p^{(n)2} \begin{pmatrix} I_1^{(n)} & I_2^{(n)} \\ I_2^{(n)} & I_3^{(n)} \end{pmatrix},\ 0,\ \sigma_{\epsilon_1}^{(n)2}\Delta t,\ \ldots,\ \sigma_{\epsilon_{M^{(n)}}}^{(n)2}\,\Delta t\Big\}. \tag{45}$$

Then, assembling over all pulsars,

$$Q_{xx} = \text{diag}\Big\{Q_{xx}^{(1)}, Q_{xx}^{(2)}, \ldots, Q_{xx}^{(N)}\Big\}. \tag{46}$$

(c) Cross-covariance between a and x_n: The cross-covariance arises from the coupling through C_n. For pulsar n we already computed the coupling block in the state-transition matrix as

$$F_{x_n,a} = \begin{pmatrix} 0 \\ 0 \\ \dfrac{1 - e^{-\gamma_a \Delta t}}{\gamma_a} \\ 0 \\ \vdots \\ 0 \end{pmatrix}. \tag{47}$$

Since the noise in a is injected directly with covariance Q_a, the cross-covariances are

$$Q_{x_n,a} = F_{x_n,a}\, Q_a \quad \text{and} \quad Q_{a,x_n} = \left(Q_{x_n,a}\right)^T. \tag{48}$$

(d) Full Process Noise Covariance: Collecting the above results, the full process noise covariance is

$$\boxed{Q(\Delta t) = \begin{pmatrix} \dfrac{1 - e^{-2\gamma_a \Delta t}}{2\gamma_a} Q_a & \begin{matrix} \vdots \\ \{F_{x_n,a}\, Q_a\}_{n=1} \\ \vdots \end{matrix} \\ \cdots & \\ \begin{matrix} Q_{a,x_1} \\ \vdots \\ Q_{a,x_N} \end{matrix} & \operatorname{diag}\left\{Q_{xx}^{(1)}, Q_{xx}^{(2)}, \ldots, Q_{xx}^{(N)}\right\} \end{pmatrix}.} \tag{49}$$

$$I_3 = \int_0^{\Delta t} e^{-2\gamma_p s}\, ds = \frac{1 - e^{-2\gamma_p \Delta t}}{2\gamma_p}. \tag{50}$$

$$I_2 = \int_0^{\Delta t} \frac{1 - e^{-\gamma_p s}}{\gamma_p} e^{-\gamma_p s}\, ds = \frac{1}{\gamma_p}\left[\int_0^{\Delta t} e^{-\gamma_p s}\, ds - \int_0^{\Delta t} e^{-2\gamma_p s}\, ds\right]. \tag{51}$$

Evaluating the two integrals separately,

$$\int_0^{\Delta t} e^{-\gamma_p s}\, ds = \frac{1 - e^{-\gamma_p \Delta t}}{\gamma_p}, \tag{52}$$

$$\int_0^{\Delta t} e^{-2\gamma_p s}\, ds = \frac{1 - e^{-2\gamma_p \Delta t}}{2\gamma_p}, \tag{53}$$

so that

$$I_2 = \frac{1}{\gamma_p}\left[\frac{1 - e^{-\gamma_p \Delta t}}{\gamma_p} - \frac{1 - e^{-2\gamma_p \Delta t}}{2\gamma_p}\right] = \frac{2\,(1 - e^{-\gamma_p \Delta t}) - (1 - e^{-2\gamma_p \Delta t})}{2\gamma_p^2}. \tag{54}$$

Next, we evaluate

$$I_1 = \int_0^{\Delta t} \left(\frac{1 - e^{-\gamma_p s}}{\gamma_p} \right)^2 ds = \frac{1}{\gamma_p^2} \int_0^{\Delta t} \left(1 - e^{-\gamma_p s}\right)^2 ds \,. \tag{55}$$

Expanding the square inside the integral,

$$\left(1 - e^{-\gamma_p s}\right)^2 = 1 - 2\, e^{-\gamma_p s} + e^{-2\gamma_p s} \,, \tag{56}$$

so that

$$I_1 = \frac{1}{\gamma_p^2} \left[\int_0^{\Delta t} 1 \, ds - 2 \int_0^{\Delta t} e^{-\gamma_p s} \, ds + \int_0^{\Delta t} e^{-2\gamma_p s} \, ds \right] . \tag{57}$$

Substituting the previously evaluated integrals, we have

$$I_1 = \frac{1}{\gamma_p^2} \left[\Delta t - 2 \, \frac{1 - e^{-\gamma_p \Delta t}}{\gamma_p} + \frac{1 - e^{-2\gamma_p \Delta t}}{2\gamma_p} \right] . \tag{58}$$

$$\boxed{I_1 = \frac{1}{\gamma_p^2} \left[\Delta t - \frac{2(1 - e^{-\gamma_p \Delta t})}{\gamma_p} + \frac{1 - e^{-2\gamma_p \Delta t}}{2\gamma_p} \right] ,} \tag{59}$$

$$\boxed{I_2 = \frac{2\left(1 - e^{-\gamma_p \Delta t}\right) - \left(1 - e^{-2\gamma_p \Delta t}\right)}{2\gamma_p^2} \,,} \tag{60}$$

$$\boxed{I_3 = \frac{1 - e^{-2\gamma_p \Delta t}}{2\gamma_p} \,.} \tag{61}$$